

Response Letter with Annotations

Zhou, A., Capizzo, L. W., Page, T. G., & Toth, E. L. (2023). Exploring public relations research topics and inter-cluster dynamics through computational modeling (2010-2020): A study based on two SSCI journals. *Journal of Public Relations Research*, 35(3), 135-161.

1. Purposes

a. Motivation:

To identify evolution and intersection of paradigms in public relations scholarship in the last decade (2010-2020), overcoming limitations of previous research that only focused on differences among paradigms and emphasized discrete research topics and methods.

b. Specific Rationale (several bullet points):

-To understand the evolution of public relations scholarship and its research clusters and paradigms, one must begin by examining how scholars have shifted directions as well as how they have reengaged with existing research questions, methods, and topics through innovative and intersecting approaches.)

-The characteristics of public relations research, researcher: broader multi-methodological expertise, varieties of modes of inquiry, and a larger toolbox to tackle disparate research problems.

-Paradigms and clusters: This study identifies clusters of existing scholarship, which are language-driven groupings of topics reflecting similar themes, research areas, theoretical frameworks, and/or methods. They admit that clusters are not representative of a true paradigm (Curtin, 2012; Kuhn, 2012), and the inherent characteristics of their identified clusters make them proto-paradigmatic

c. Research Questions and/or Hypotheses

RQ1: How, if at all, have public relations research topics (2010–2020) clustered together?

RQ2: Between 2010 and 2020, how have public relations research clusters' proportions differed across journals?

RQ3: How have public relations research clusters intersected with each other in the past decade (RQ3a), and what is the degree of interconnectedness among these existing clusters (RQ3b)?

-Journals: Public Relations Review (PRR) and the Journal of Public Relations Research (JPRR)

-Public relations research clusters: paradigm-like groupings of articles with significant topical connectivity uncovered through structural topic modeling, thematic analysis, and inter-cluster network analysis.

2. Method

a. Research Design

Contents analysis (topic modeling) and network analysis

b. Sample

Public Relations Review (PRR) and the Journal of Public Relations Research (JPRR), two SSCI-indexed journals focusing on public relations research

Data collection: Scraping of every article that was published between 2010 and 2020 in these journals. The ultimately obtained dataset contains 1,093 articles from PRR and 200 articles from JPRR, as well as each article's metadata (e.g., year, authors, keywords, abstract, and number of citations). They collected each article's full text, which amounts to 7,400,685 words in total.

c. Measures

d. Analysis

The paper conducted analyses on a textual corpus they obtained from the sample with three computational methods: structural topic modeling (including qualitative or thematic analysis), inter-cluster network analysis, and network simulation.

-Structural topic modeling (STM):

STM incorporates covariates (year and journal) and examines their impact on topical proportion. They selected 65, considering both model diagnosis (such as semantic coherence and residuals) by estimating the number of topics from 20 to 150 and qualitative reading, following best practices (Song et al., 2020; Zhou, 2021),

-Inter-cluster network analysis:

The study treats each article as a combination of research clusters and uses 1,293 articles to construct a weighted inter-cluster network that shows how clusters co-occur. This network is analyzed to identify patterns of inter-cluster development in public relations research, with the Gini coefficient used to assess preferential overlap between clusters.

-Network simulation:

They used the bipartite package in R, *decimalr2dtable* function. The study simulated 1000 "parallel universes," constructed an inter-cluster network for each matrix, and calculated the tie strengths of these 1000 networks. The 95% and 5% bounds of these tie strengths are then used as confidence intervals to determine whether observed inter-cluster contributions significantly differ from those expected under null models.

3. Results and Conclusions

a. Answers to RQs and Hypotheses

RQ1: Clusters in Public Relations Research, 2010–2020

Answer:

Result suggests 9 research clusters: the strategic management cluster emerged as the most central for the past decade, followed by public relations professionalism, digital media, crisis communication, internal communication, global public relations, rhetoric and philosophy, media relations, and critical studies, ranked by their proportions in the scholarship

RQ2: Research Cluster Proportion Shift from 2010 to 2020 across Journals

Answer:

- Little longitudinal change occurred throughout the decade
- JPRR had greater emphasis on the strategic management cluster relative to PRR, which offered a more diverse representation.

RQ3a: Inter-Cluster Knowledge Production

Answer:

The strategic management research cluster is the most influential in terms of inter-cluster exposure, occupying the “hub” position in the network, thereby facilitating inter-cluster scholarship; however, other research areas are disproportionately dependent on it.

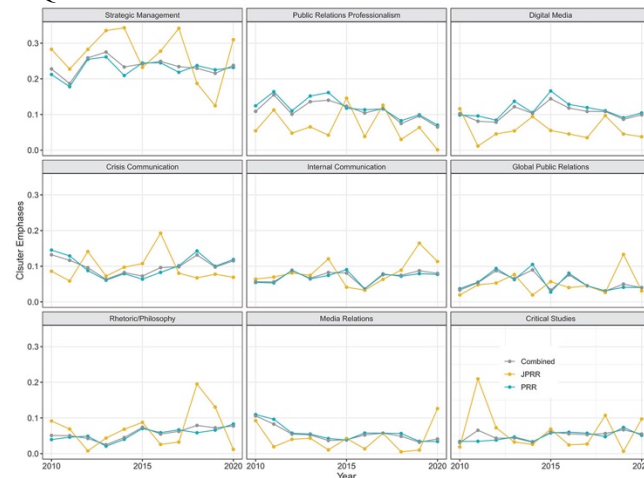
RQ3b: Inter-Cluster Network Simulation

Answer:

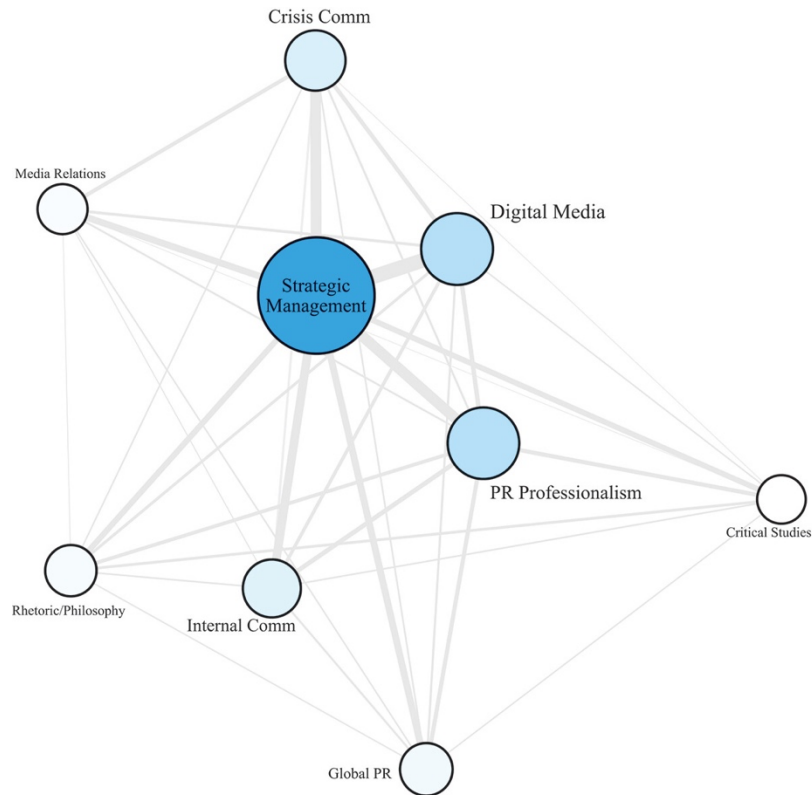
The last ten years of public relations research have not seen as much intersection among theoretical traditions across clusters as expected: Across all 36 ties, 31 are significantly weaker than simulation, indicating that the landscape of public relations research has much less inter-cluster knowledge production than it could have.

b. Include Key Figures/Tables

RQ 2:



RQ3a



RQ3b

Table 4. Inter-cluster network adjacency matrix and simulation results.

	Public Relations Professionalism	Digital Media	Crisis Communication	Internal Communication	Global Public Relations	Rhetoric and Philosophy	Media Relations	Critical Studies
Strategic Management	20.549 [26.535, 26.615]	28.031 [25.162, 25.240]	20.493 [23.409, 23.483]	16.518 [18.079, 18.139]	12.960 [14.895, 14.946]	11.071 [14.593, 14.644]	12.801 [14.059, 14.110]	10.528 [13.308, 13.355]
Public Relations Professionalism		8.494 [13.193, 13.244]	4.114 [12.235, 12.286]	8.630 [9.337, 9.377]	7.934 [7.661, 7.696]	6.890 [7.490, 7.523]	3.735 [7.185, 7.218]	7.889 [6.823, 6.855]
Digital Media			7.872 [11.559, 11.607]	6.484 [8.755, 8.793]	4.181 [7.168, 7.200]	5.363 [7.036, 7.068]	5.337 [6.742, 6.774]	2.900 [6.377, 6.406]
Crisis Communication				4.440 [8.144, 8.180]	3.236 [6.648, 6.680]	3.048 [6.521, 6.552]	7.374 [6.293, 6.323]	1.411 [5.910, 5.938]
Internal Communication					4.044 [4.997, 5.022]	3.107 [4.890, 4.915]	2.110 [4.698, 4.723]	3.210 [4.441, 4.464]
Global Public Relations						2.664 [3.981, 4.002]	3.071 [3.832, 3.853]	2.714 [3.612, 3.632]
Rhetoric and Philosophy							1.579 [3.751, 3.772]	5.239 [3.558, 3.579]
Media Relations								1.035 [3.402, 3.421]

Note: Bold text indicates that the observed tie between two clusters is significantly stronger than the simulated parallel universes. Italic text indicates that the observed tie between two clusters is significantly weaker than simulation. The table shows that the landscape of public relations research—as we the scholarly community have constructed in the past decade—has much less inter-cluster knowledge production than what would have been expected. .

c. Authors' Conclusions

By combining topic modeling and network analysis, this study offers a clear and comprehensive representation of public relations scholarship published between 2010 and

2020 in two leading journals, PRR and the JPRR. This comprehensive dataset and analytical approach deepen understanding of connections and gaps across individual research trajectories and the field as a whole, highlighting patterns of both refraction and recommitment in public relations research.

4. Discussion

a. Implications

This comprehensive dataset and analytical approach enhance understanding of how individual research trajectories and the field as a whole are connected, or disconnected, highlighting patterns of both divergence and continuity in public relations scholarship. More broadly, the study offers a foundation for a new discourse on convergence and integration across communication, management, and mass communication research and practice.

b. Limitations

By selecting only the two public relations-oriented SSCI journals, it does not capture the full scope of diversity in the discipline, so resulting research clusters and overall implications are bounded by these two journals.

c. Future Directions

Future research may expand this work by incorporating additional journals and adopting a longitudinal perspective to build on these findings.

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Response Letter (~600 words)

I believe this originality is significant in multiple ways. The research gap and questions they raised are valid and valuable particularly for public relations discipline. Their approach to understand the public relations scholarship, which tries to show how studies in this area evolved and are interconnected to each other, beyond identifying what has been studied and differences among them, is reasonable and appropriate considering the characteristics of public relations scholarly discipline. I like that they explained characteristics of public relations researchers to strengthen their rationales on their approach.

Furthermore, computational methods they used were rigorous. I especially liked that computational methods themselves are means, not a goal for this work. Even if I had little knowledge on three main computational methods of this work before I read it, this paper helped me understand functions and roles of three methods because they were used to answer the research questions, and the paper explained in detail how they used it to address research questions. I think methods were very rigorous as well. For example, although clusters in this paper are the same as paradigms and it's a limitation, how they used STM to form clusters that can be proxy to paradigms helped me understand what STM can do. Their descriptions of their qualitative works during STM process and

process such as omitting method clusters emphasized how their paradigm-like cluster approach is different from approach of previous research categorizing works by topics or methods.

This work suggests sample representativeness as its weakness. They explained that because they included only two journals, their research clusters and implications are bounded by this journal sample. Although this limitation is undeniable, I don't think it is a critical issue. No journal sample can be neutral or representative and including more journals or even all journals are not only methodologically challenging but also not meaningful. Rather, I consider the lack of explanations about these two journals a weakness. I think it would be more helpful if they explained why they selected these two journals and what are their characteristics regarding interpretation of results. They mentioned that they are two SSCI journals in public relations and suggested that these are prominent journals. However, it would be helpful to provide more explanation of each journal's scope and selection criteria in relation to the research questions. A more detailed explanation, such as the journal's original purpose, how it articulates its core values, and what distinguishes it from other journals based on objectively established information, would make it easier to understand the significance of changes over time and the interconnections among different research clusters. Most importantly, lack of information on these two journals made me think the rationale for comparing the two journals is less convincing. Although the journals are compared, if their identities are not clearly understood or the reason for comparing them is not explicitly justified, the results could feel like descriptive information without meaningful analytical value. I think cluster comparisons between conservative newspapers and liberal newspapers could benefit so much as it informs us meaningful information following this paper.

In conclusion, I consider this study to be an excellent paper as it offers significant value across three key dimensions: the strength of its intellectual motivation and research questions, the rigor of its methodological approach, and its implications. In particular, its implications are substantial as it goes beyond conventional approaches in research review scholarship in public relations by proposing a new perspective, and by suggesting the need for greater inter-cluster knowledge production as a meaningful outcome of its findings.